Supplementary Materials

1 Hinge Kinematics for Robot Base Planner

1.1 Hinge Kinematics

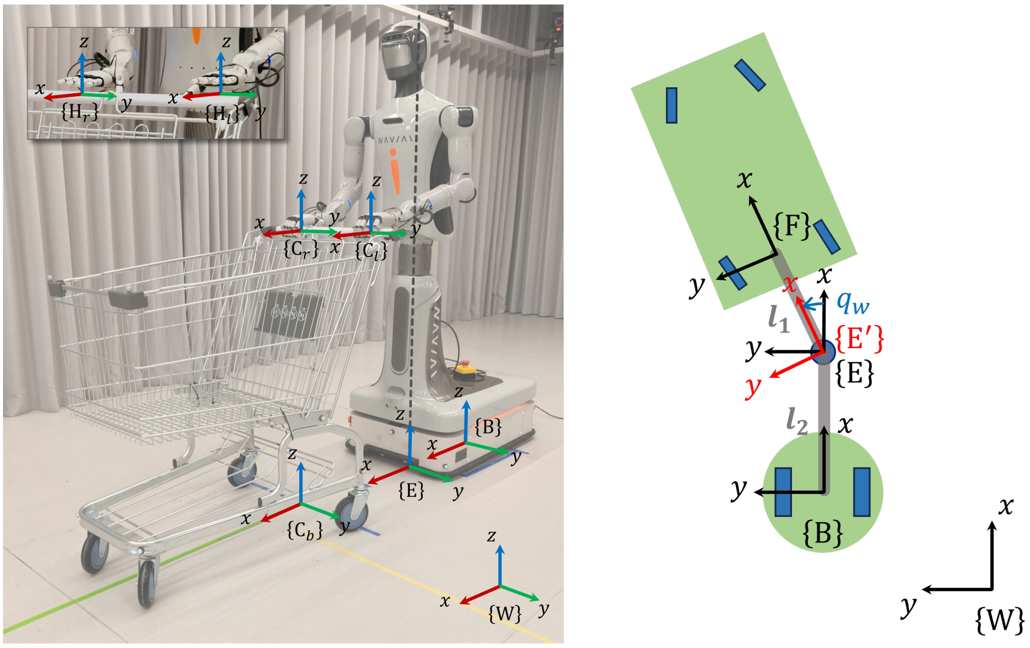


Fig. 1. Frame definitions for the cart-handling system.

Hinge kinematics is used to plan the motion of the robot chassis during cart handling tasks. As shown on the right side of Fig. 1, given the cart state $(\boldsymbol{x}_{\mathrm{F}}, \boldsymbol{v}_{\mathrm{F}})$ in {W} and the waist-yaw angle $q_w, \dot{q}_w$, the chassis state $(\boldsymbol{x}_{\mathrm{B}}, \boldsymbol{v}_{\mathrm{B}})$ follows

$$\begin{bmatrix} x_{\mathrm{B}} \\ y_{\mathrm{B}} \\ \theta_{\mathrm{B}} \end{bmatrix} = \begin{bmatrix} x_{\mathrm{F}} \\ y_{\mathrm{F}} \\ \theta_{\mathrm{F}} \end{bmatrix} + \begin{bmatrix} -l_1 c\theta_{\mathrm{F}} - l_2 c(\theta_{\mathrm{F}} - q_w) \\ -l_1 s\theta_{\mathrm{F}} - l_2 s(\theta_{\mathrm{F}} - q_w) \\ -q_w \end{bmatrix}, \tag{1}$$

$$\begin{bmatrix} \dot{x}_{\mathrm{B}} \\ \dot{y}_{\mathrm{B}} \\ \dot{\theta}_{\mathrm{B}} \end{bmatrix} = \begin{bmatrix} \dot{x}_{\mathrm{F}} \\ \dot{y}_{\mathrm{F}} \\ \dot{\theta}_{\mathrm{F}} \end{bmatrix} + \begin{bmatrix} l_1 s\theta_{\mathrm{F}} + l_2 s(\theta_{\mathrm{F}} - q_w) & l_2 s(\theta_{\mathrm{F}} - q_w) \\ -l_1 c\theta_{\mathrm{F}} - l_2 c(\theta_{\mathrm{F}} - q_w) & -l_2 c(\theta_{\mathrm{F}} - q_w) \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_{\mathrm{F}} \\ \dot{q}_w \end{bmatrix}, \tag{2}$$

where $l_1 = |\mathrm{FE}|, l_2 = |\mathrm{EB}|$, and $c\theta = \cos\theta, s\theta = \sin\theta$. We compactly write the hinge kinematics as $(\boldsymbol{x}_{\mathrm{B}}, \boldsymbol{v}_{\mathrm{B}}) = \mathrm{HK}(\boldsymbol{x}_{\mathrm{F}}, \boldsymbol{v}_{\mathrm{F}}, q_w, \dot{q}_w)$.

1.2 Robot Base Planner

Before whole-body control, a reference trajectory of the robot base and waist-yaw needs to be generated under the hinge kinematics and the nonholonomic constraint of the robot differential-drive wheels. This robot base planner helps coordinate the movement of the chassis and upper body, and prevents the upper body from reaching singular or infeasible configurations. Treating $\boldsymbol{x}_{\mathrm{E}'}$ (related to the current cart pose $\boldsymbol{x}_{\mathrm{F}}$) as the reference waist state, we optimize $\boldsymbol{x}_{\mathrm{B}}$ and q_w to make $\boldsymbol{x}_{\mathrm{E}}$ move toward $\boldsymbol{x}_{\mathrm{E}'}$ as fast as possible while satisfying constraints. The reference waist angle and velocity are computed as follows:

$$q_w^{ref} = \mathrm{atan2}\left(\overrightarrow{\mathrm{BE}'} \times \overrightarrow{\mathrm{E}'\mathrm{F}}, \overrightarrow{\mathrm{BE}'} \cdot \overrightarrow{\mathrm{E}'\mathrm{F}}\right), \dot{q}_w^{ref} = \omega_{x_{\mathrm{F}}} - \omega_{x_{\mathrm{B}}}. \tag{3}$$

We define the state and input of this reference generator as $\mathbf{x} = [\boldsymbol{x}_{\mathrm{B}}, q_w, \boldsymbol{v}_{\mathrm{B}}, \dot{q}_w]^T$ and $\mathbf{u} = [\ddot{\boldsymbol{x}}_{\mathrm{B}}, \ddot{q}_w]^T$. The problem is then formulated as

$$\min_{\mathbf{u}(\cdot)} J_1 = \int_{t=t_0}^{t_0+T} L_1\left(\mathbf{x}(t), \mathbf{x}^{ref}(t), \mathbf{u}(t)\right)\, dt, \tag{4}$$

$$\text{s.t.} \quad \dot{\mathbf{x}}(t) = \begin{bmatrix} \mathbf{0} & \mathbf{I_4} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} \mathbf{0} \\ \mathbf{I_4} \end{bmatrix} \mathbf{u}(t), \tag{5}$$

$$\mathbf{x}^{ref}(t) = \mathrm{HK}\left(\boldsymbol{x}_{\mathrm{F}}(t), \boldsymbol{v}_{\mathrm{F}}(t), q_w^{ref}(t), \dot{q}_w^{ref}(t)\right), \tag{6}$$

$$-q_{w,max} \leq q_w(t) \leq q_{w,max}, \tag{7}$$

$$-\mathbf{v}_{max} \leq [\boldsymbol{v}_{\mathrm{B}}(t), \dot{q}_w(t)] \leq \mathbf{v}_{max}, \tag{8}$$

$$-\mathbf{u}_{max} \leq \mathbf{u}(t) \leq \mathbf{u}_{max}, \tag{9}$$

$$\left|\boldsymbol{v}_{y_{\mathrm{B}}} \cos\theta_{\mathrm{B}} - \boldsymbol{v}_{x_{\mathrm{B}}} \sin\theta_{\mathrm{B}}\right| \leq \epsilon, \tag{10}$$

where $\mathrm{HK}(\cdot)$ denotes the hinge kinematic mapping, and the cost function is defined as

$$\begin{aligned} L_1 = {} & \left(\mathbf{x} - \mathbf{x}^{ref}\right)^T \mathbf{Q}_1 \left(\mathbf{x} - \mathbf{x}^{ref}\right) + \mathbf{u}^T \mathbf{R}_1 \mathbf{u} \\ & + \mathrm{K}_1 \left(\boldsymbol{v}_{y_{\mathrm{B}}} \cos\theta_{\mathrm{B}} - \boldsymbol{v}_{x_{\mathrm{B}}} \sin\theta_{\mathrm{B}}\right)^2 + \mathrm{K}_2 q_w^2 + \mathrm{K}_3 \dot{q}_w^2. \end{aligned} \tag{11}$$

With this optimization, the base and waist-yaw references ($\boldsymbol{x}_{\mathrm{B}}^{des}, \boldsymbol{v}_{\mathrm{B}}^{des}, q_w^{des}, \dot{q}_w^{des}$) can be solved online while satisfying kinematic constraints. These references are then sent to a whole-body MPC controller to optimize the motion of the entire robot.

2 Parameters for RL Training

2.1 Sampling Probabilities for Motion Mode Commands

The initial probability distribution of each motion mode is $P_0(m), m \in M_0$, whose values are listed in Table I.

TABLE I
SAMPLING PROBABILITIES FOR MOTION MODE COMMANDS.

Description	Probability
Motion Type	
only straight	$p(m = a) = 0.50$
only turning	$p(m = b) = 0.05$
straight and turning	$p(m = c) = 0.35$
standing	$p(m = d) = 0.10$
Motion Heading	
forward	$p(v > 0) = 0.50$
leftward	$p(\omega > 0) = 0.50$

2.2 Observation Noise

The observation space and additive noise settings are listed in Table II.

2.3 Termination Thresholds

Five termination conditions are set: episode timeout, cart too lifted, cart too tilted, EEs too far from the cart handle, and EEs too far from the nominal trajectories. The thresholds are listed in Table III.

TABLE II

Observation space and additive noise for the actor $(\boldsymbol{o}_t)$ and critic $(\boldsymbol{o}_t, \boldsymbol{o}_t^{pr})$. $k \in \{l, r\}$ denotes the left/right EE. $X_{\mathrm{B}}^{\mathrm{A}}$ denotes the error of frame B relative to frame A, expressed in frame B.

Description	Symbol	Dim.	Noise
	$\boldsymbol{o}_t$		
Cart lifting	${}^{w}p_{\mathrm{C}_b}$	1	$U(\pm 0.001)$
Cart tilting	${}^{w}\psi_{\mathrm{C}_b}$	1	$U(\pm 0.001)$
Cart twist	${}^{w}\boldsymbol{\mathcal{V}}_{\mathrm{C}_b}$	6	$U(\pm 0.005)$
Cart and EE waypoints	${}^{w}\boldsymbol{x}_{\mathrm{C}_b}^{nomN}, {}^{w}\boldsymbol{x}_{\mathrm{C}_k}^{nomN}, {}^{b}\boldsymbol{v}_{\mathrm{C}_b}^{nomN}, {}^{b}\boldsymbol{v}_{\mathrm{C}_k}^{nomN}$	180	–
Cart and EE tracking error	${}^{b}\boldsymbol{x}_{\mathrm{C}_b}^{\mathrm{C}_b nom}, {}^{b}\boldsymbol{\mathcal{X}}_{\mathrm{H}_k}^{\mathrm{C}_k nom}, {}^{b}\boldsymbol{v}_{\mathrm{C}_b}^{\mathrm{C}_b nom}, {}^{b}\boldsymbol{\mathcal{V}}_{\mathrm{H}_k}^{\mathrm{C}_k nom}$	30	$U(\pm 0.005)$
EE control error	${}^{b}\boldsymbol{\mathcal{X}}_{\mathrm{H}_k}^{\mathrm{H}_k des}, {}^{b}\boldsymbol{\mathcal{V}}_{\mathrm{H}_k}^{\mathrm{H}_k des}$	24	$U(\pm 0.005)$
EE twist	${}^{w}\boldsymbol{\mathcal{V}}_{\mathrm{H}_k}$	12	$U(\pm 0.005)$
EE joint	q_k	2	$U(\pm 0.010)$
EE pose fetching error	${}^{b}\boldsymbol{\mathcal{X}}_{\mathrm{H}_k}^{\mathrm{C}_k}$	12	$U(\pm 0.002)$
Action buffer	$\boldsymbol{a}_t, \boldsymbol{a}_{t-1}, \boldsymbol{a}_{t-2}$	42	–
	$\boldsymbol{o}_t^{pr}$		
EE twist fetching error	${}^{b}\boldsymbol{\mathcal{V}}_{\mathrm{H}_k}^{\mathrm{C}_k}$	12	–
Measured wrench at wrists	${}^{b}\boldsymbol{\mathcal{W}}_{\mathrm{H}_k}$	12	–
Cart mass	m_{C}	1	–

TABLE III

Termination thresholds.

Description	Threshold
Episode timeout	$t_E = 30\,\mathrm{s}$
Cart lifted	$p_{\varepsilon_0} = 0.05\,\mathrm{m}$
Cart tilted	$\psi_{\varepsilon_0} = 0.05\,\mathrm{rad}$
EE fetching error (pos & ori)	$\boldsymbol{\rho}_{\varepsilon_0} = [0.04, 0.16, 0.03]\,\mathrm{m}$, $\boldsymbol{\phi}_{\varepsilon_0} = [0.20, 0.20, 0.40]\,\mathrm{rad}$
EE tracking error (pos & ori)	$\rho_{\varepsilon_0} = 0.20\,\mathrm{m}$, $\phi_{\varepsilon_0} = 0.40\,\mathrm{rad}$

2.4 Reward Weights

Table IV lists all reward terms, formulations, and weights.

2.5 Domain Randomization

Domain randomization (DR) is adopted during RL training to improve robustness. The parameter ranges are summarized in Table V, where the force and torque directions are generated by normalizing Gaussian random vectors to unit length when applying external disturbances.

3 Simulation Results for Different End-Effectors

In addition to dexterous hands, we evaluate the baselines with three-finger grippers and ring-shaped EEs in IsaacLab simulation. The two EE geometries are shown in Fig. 2, and the results are summarized in Tables VI and VII.

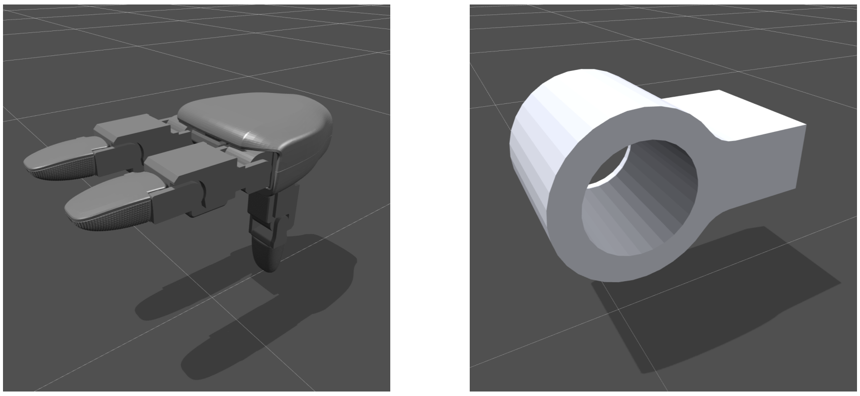
TABLE IV

REWARDS. $i \in \{1, \dots, 4\}$ AND $k \in \{l, r\}$.

Description	Formulation	Weight
	Task Rewards	
Cart pose tracking	$r_{1,t} = w_1 \exp\left(-\sum_{i=1}^{4} \left\| {}^{w}\boldsymbol{x}_{\mathrm{C}_i}^{nom} - {}^{w}\boldsymbol{x}_{\mathrm{C}_i} \right\|_2^2 / \sigma_1^2\right)$	$w_1 = 20, \sigma_1 = 0.05$
Cart twist tracking	$r_{2,t} = w_2 \exp\left(-\sum_{i=1}^{4} \left\| {}^{b}\boldsymbol{v}_{\mathrm{C}_i}^{nom} - {}^{b}\boldsymbol{v}_{\mathrm{C}_i} \right\|_2^2 / \sigma_2^2\right)$	$w_2 = 10, \sigma_2 = 0.05$
EE pose tracking	$r_{3,t} = w_3 \sum_{k=l,r} \left[\exp\left(-\left\|\mathbf{K}_{pos}\cdot {}^{b}\boldsymbol{\rho}_{\mathrm{H}_k}^{\mathrm{C}_k nom}\right\|_2^2/\sigma_3^2\right)\right.$ $\left. + \exp\left(-\left\|\mathbf{K}_{ori}\cdot {}^{b}\boldsymbol{\phi}_{\mathrm{H}_k}^{\mathrm{C}_k nom}\right\|_2^2/\sigma_3^2\right)\right]$	$w_3 = 5, \sigma_3 = 0.05$ $\mathbf{K}_{pos} = \mathbf{K}_{ori} = [2.0, 1.5, 2.0]$
EE twist tracking	$r_{4,t} = w_4 \sum_{k=l,r} \left[\exp\left(-\left\|\mathbf{K}_{lin}\cdot {}^{b}\boldsymbol{v}_{\mathrm{H}_k}^{\mathrm{C}_k nom}\right\|_2^2/\sigma_4^2\right)\right.$ $\left. + \exp\left(-\left\|\mathbf{K}_{ang}\cdot {}^{b}\boldsymbol{\omega}_{\mathrm{H}_k}^{\mathrm{C}_k nom}\right\|_2^2/\sigma_4^2\right)\right]$	$w_4 = 5, \sigma_4 = 0.05$ $\mathbf{K}_{lin} = \mathbf{K}_{ang} = [2.0, 1.5, 2.0]$
EE pose fetching	$r_{5,t} = w_5 \sum_{k=l,r} \left[\exp\left(-\left\|\mathbf{K}_{pos}\cdot {}^{b}\boldsymbol{\rho}_{\mathrm{H}_k}^{\mathrm{C}_k}\right\|_2^2/\sigma_5^2\right)\right.$ $\left. + \exp\left(-\left\|\mathbf{K}_{ori}\cdot {}^{b}\boldsymbol{\phi}_{\mathrm{H}_k}^{\mathrm{C}_k}\right\|_2^2/\sigma_5^2\right)\right]$	$w_5 = 0.5, \sigma_5 = 1.0$ $\mathbf{K}_{pos} = \mathbf{K}_{ori} = [2.0, 1.5, 2.0]$
EE consistency	$r_{6,t} = w_6\left[\exp\left(-\left\|{}^{w}\boldsymbol{\rho}_{z_{\mathrm{H}_l}} - {}^{w}\boldsymbol{\rho}_{z_{\mathrm{H}_r}}\right\|_2^2/\sigma_6^2\right)\right.$ $\left. + \exp\left(-\left\|{}^{w}\boldsymbol{\phi}_{\mathrm{H}_l} - {}^{w}\boldsymbol{\phi}_{\mathrm{H}_r}\right\|_2^2/\sigma_6^2\right)\right]$	$w_6 = 0.5, \sigma_6 = 2.0$
EE gripping	$r_{7,t} = w_7 \sum_{k=l,r} \left[\mathrm{I}\cdot\exp((q_{open} - q_k)/\sigma_7^2)\right.$ $\left. + (1-\mathrm{I})\cdot\exp((q_k - q_{close})/\sigma_7^2)\right],$ if ${}^{b}\boldsymbol{\rho}_{\mathrm{H}_k}^{\mathrm{C}_k} > \boldsymbol{\rho}_{\varepsilon_1}$ or ${}^{b}\boldsymbol{\phi}_{\mathrm{H}_k}^{\mathrm{C}_k} > \boldsymbol{\phi}_{\varepsilon_1}, \mathrm{I} = 1;$ else $\mathrm{I} = 0.$	$w_7 = 0.5, \sigma_7 = 0.5$ $\boldsymbol{\rho}_{\varepsilon_1} = [0.03, 0.12, 0.02]$ $\boldsymbol{\phi}_{\varepsilon_1} = [0.15, 0.15, 0.30]$ $q_{open} = 0.0$ rad, $q_{close} = 1.0$ rad
	Balance Penalties	
Cart lifted	$r_{8,t} = w_8 \left({}^{w}p_{\mathrm{C}_b}\right)^2$	$w_8 = -2\mathrm{e}5$
Cart tilted	$r_{9,t} = w_9 \left({}^{w}\psi_{\mathrm{C}_b}\right)^2$	$w_9 = -1\mathrm{e}5$
Cart sliding	$r_{10,t} = w_{10} \left({}^{b}\boldsymbol{v}_{y_{\mathrm{C}_b}}\right)^2$	$w_{10} = -1\mathrm{e}3$
Cart undesired twist	$r_{11,t} = w_{11} \left[\left({}^{b}\boldsymbol{\omega}_{X_{\mathrm{C}_b}}\right)^2 + \left({}^{b}\boldsymbol{\omega}_{Y_{\mathrm{C}_b}}\right)^2\right]$	$w_{11} = -1\mathrm{e}3$
EE undesired twist	$r_{12,t} = w_{12} \sum_{k=l,r}\left[\left({}^{b}\boldsymbol{\omega}_{X_{\mathrm{H}_k}}\right)^2 + \left({}^{b}\boldsymbol{\omega}_{Y_{\mathrm{H}_k}}\right)^2\right]$	$w_{12} = -1\mathrm{e}2$
	Smoothness Penalties	
Velocity smoothness	$r_{13,t} = w_{13} \sum_{k=l,r} \left\|{}^{w}\boldsymbol{\mathcal{V}}_{\mathrm{H}_k}^{des}\right\|_2^2$	$w_{13} = -1.0$
Acceleration smoothness	$r_{14,t} = w_{14} \sum_{k=l,r} \left\|{}^{w}\boldsymbol{\mathcal{A}}_{\mathrm{H}_k}^{des}\right\|_2^2$	$w_{14} = -1\mathrm{e}{-2}$
	Survival Bonus	
Milestone bonus	$r_{15,t} = \mathrm{bonus}_i,$ if survival span t first reaches $\mathrm{milestone}_i$, $\mathrm{milestone}_i \in [0, 50, 150, 300, 450, 600, 750]$, $\mathrm{bonus}_i \in [0, 1, 5, 10, 20, 30, 45]$.	$w_{15} = 1.0$

TABLE V
DOMAIN RANDOMIZATION RANGES.

Description	Easy	Hard
Friction coefficient ($\mu_s \geq \mu_d$)	$\mu_s \sim U(0.7, 0.9), \mu_d \sim U(0.5, 0.7)$	$\mu_s \sim U(0.6, 1.0), \mu_d \sim U(0.4, 0.8)$
Caster steering-axle and rolling-axle damping	$\eta_s \sim U(0.02, 0.08), \eta_r \sim U(0.01, 0.04)$	$\eta_s \sim U(0.02, 0.20), \eta_r \sim U(0.01, 0.10)$
Floating EE controller PD scaling factor	$\mathrm{K}_{dr} \sim U(0.9, 1.1)$	$\mathrm{K}_{dr} \sim U(0.8, 1.2)$
Cart mass scaling factor	$\mathrm{K}_{\mathrm{C}} \sim U(1.0, 1.5)$	$\mathrm{K}_{\mathrm{C}} \sim U(1.0, 2.0)$
Cart CoM position offset factor	$\mathrm{K}_{\Delta} \sim U(-0.1, 0.1), (\Delta = \mathrm{K}_{\Delta}[l_x, l_y, l_z])$	$\mathrm{K}_{\Delta} \sim U(-0.2, 0.2), (\Delta = \mathrm{K}_{\Delta}[l_x, l_y, l_z])$
Caster steering-axle initial angle	$q_s \sim U(-\pi/4, \pi/4)$	$q_s \sim U(-\pi, \pi)$
EE initial pose offset w.r.t. the cart handle	$\delta\boldsymbol{\rho}_y \sim U(-0.05, 0.05)$	$\delta\boldsymbol{\rho}_y \sim U(-0.05, 0.05), \delta\boldsymbol{\rho}_x, \delta\boldsymbol{\rho}_z \sim U(-0.005, 0.005)$ $\delta\boldsymbol{\phi}_X, \delta\boldsymbol{\phi}_Y, \delta\boldsymbol{\phi}_Z \sim U(-0.025, 0.025)$
Random external disturbances on the cart	$\lVert\boldsymbol{f}_e\rVert \sim U(0, 10), \lVert\boldsymbol{\tau}_e\rVert \sim U(0, 1.0),$ $t_d \sim U(0.1, 0.5), t_i \sim U(0, 5)$	$\lVert\boldsymbol{f}_e\rVert \sim U(0, 20), \lVert\boldsymbol{\tau}_e\rVert \sim U(0, 2.0),$ $t_d \sim U(0.1, 0.5), t_i \sim U(0, 5)$



(a) Three-finger gripper. (b) Ring-shaped EE.

Fig. 2. Additional EE configurations evaluated in IsaacLab simulation.

TABLE VI
PUSH SIMULATION RESULTS WITH DIFFERENT EE CONFIGURATIONS ON TrajA. THE BEST RESULT IN EACH COLUMN IS SHOWN IN RED.

EE Config.	Method	Push Pos Error [m]			Push Ori Error [rad]		
		MAE	MAX	Terminal	MAE	MAX	Terminal
Three-finger gripper	Open-loop	0.042±0.000	0.064±0.000	0.061±0.000	0.021±0.000	0.078±0.000	0.014±0.000
	Closed-loop	0.023±0.000	**0.038±0.000**	0.017±0.000	0.018±0.000	0.074±0.000	0.009±0.000
	SimpleRL	0.042±0.010	0.076±0.021	0.037±0.006	0.064±0.008	0.153±0.013	0.058±0.010
	Ours	**0.009±0.000**	0.064±0.001	**0.011±0.000**	**0.007±0.002**	**0.037±0.005**	**0.004±0.001**
Ring-shaped EE	Open-loop	0.042±0.000	0.068±0.000	**0.014±0.000**	0.023±0.000	0.105±0.000	0.011±0.000
	Closed-loop	0.043±0.000	0.064±0.000	0.061±0.000	0.020±0.000	0.091±0.000	**0.004±0.000**
	SimpleRL	0.059±0.001	0.113±0.001	0.024±0.004	0.046±0.000	0.091±0.001	0.061±0.004
	Ours	**0.014±0.001**	**0.021±0.001**	0.020±0.002	**0.012±0.001**	**0.039±0.005**	0.020±0.005

TABLE VII
PULL SIMULATION RESULTS WITH DIFFERENT EE CONFIGURATIONS ON TrajA. THE BEST RESULT IN EACH COLUMN IS SHOWN IN RED.

EE Config.	Method	Pull Pos Error [m]			Pull Ori Error [rad]		
		MAE	MAX	Terminal	MAE	MAX	Terminal
Three-finger gripper	Open-loop	0.050±0.000	0.072±0.000	0.059±0.000	0.009±0.000	0.031±0.000	0.002±0.000
	Closed-loop	0.052±0.000	**0.068±0.000**	0.057±0.000	0.008±0.000	**0.023±0.000**	0.006±0.000
	SimpleRL	0.060±0.003	0.113±0.004	0.051±0.003	0.053±0.003	0.194±0.008	0.021±0.027
	Ours	**0.011±0.001**	0.069±0.003	**0.006±0.000**	**0.006±0.001**	0.044±0.005	**0.001±0.002**
Ring-shaped EE	Open-loop	0.051±0.000	0.077±0.000	0.026±0.000	0.011±0.000	0.043±0.000	0.007±0.000
	Closed-loop	0.050±0.000	0.071±0.000	0.058±0.000	0.011±0.000	0.037±0.000	**0.001±0.000**
	SimpleRL	0.056±0.001	0.090±0.003	0.031±0.002	0.048±0.002	0.097±0.003	0.087±0.007
	Ours	**0.014±0.000**	**0.023±0.000**	**0.005±0.003**	**0.007±0.000**	**0.036±0.003**	0.022±0.007